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A luminescent $Cu_2I_2P_2S_2$-type binuclear complex and its fluorescence sensing for pyridine

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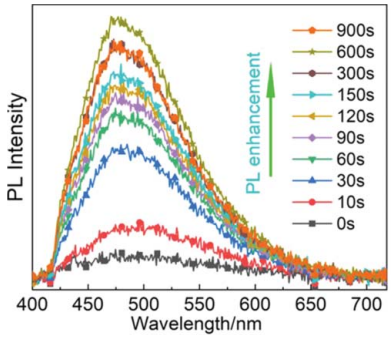
Luminescent Cu^I complexes are an important class of coordination compounds due to their relative abundance, low cost and ability to display excellent luminescence. The title $Cu_2I_2P_2S_2$-type binuclear complex, di-μ-iodido-bis[(thiourea-κS)(triphenylphosphine-κP)copper(I)], $[Cu_2I_2(CH_4N_2S)_2(C_{18}H_{15}P)_2]$, conventionally abbreviated as $Cu_2I_2TPP_2TU_2$, where TPP and TU represent triphenylphosphine and thiourea, respectively, is described. In this complex, each Cu^I atom adopts a CuI_2PS four-coordination mode and pairs of atoms are connected to each other by two μ_2-I ligands to form a centrosymmetric binuclear cluster. It was also found that the paper-based film of this complex exhibited obvious luminescence light-up sensing for pyridine and 4-methylpyridine.

1. Introduction

For decades, luminescent copper(I) complexes have attracted much attention because of their interesting photophysical properties (Liu *et al.*, 2018; Huitorel *et al.*, 2018; Chen *et al.*, 2018; Cariati *et al.*, 2016; Czerwieniec *et al.*, 2016). Organic light-emitting diodes (OLEDs) are a rapidly developing emerging display technology, and the application of Ir^{III} complexes as the core luminescent layer material has achieved great success (Chai *et al.*, 2015; Czerwieniec *et al.*, 2011, 2016; Wang *et al.*, 2020*a*; Volz *et al.*, 2015). Due to the advantages of copper, such as low price, environmental friendliness and abundance, Cu^I phosphors are considered a promising substitute material for noble Ir^{III} complexes (Volz *et al.*, 2015; Xiao *et al.*, 2011). So far, a large number of different structural types of Cu^I complexes have been designed and explored, such as the mononuclear three-coordinate type (Hashimoto *et al.*, 2011; Hofbeck *et al.*, 2015), the Cu*X*NP_2 type (*X* is a halogen atom) (Ohara *et al.*, 2014), the $[Cu(N^\wedge N)P_2]^+$ heteroleptic type (Li *et al.*, 2017; Czerwieniec *et al.*, 2011), the binuclear Cu_2X_2 cluster type (Liang *et al.*, 2018; Volz *et al.*, 2013), the ligand-bridged polycentric or polymeric type (Lin *et al.*, 2017; Kobayashi *et al.*, 2020), and so on (Tsuboyama, 2019; Troyano *et al.*, 2021). In recent decades, extensive research has been carried out on the sensing and detection of luminescence chromic and quenching modes (Kondo *et al.*, 2020; Wang *et al.*, 2022*a*,*b*). Compared with these methods, the light-up mode (luminescence enhancement from darkness) can be easily observed with the naked eye, which is convenient for sensing and detection applications. The light-up mode can be used for real-time *in situ* visual detection of target molecules, but so far, it is still challenging to design light-emitting sensors.



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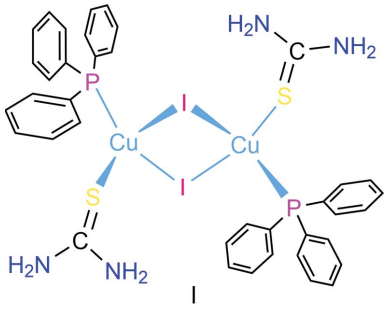
Recently, we reported that a three-coordinate type and two binuclear Cu^I clusters exhibited a luminescence enhancement response to pyridine (Wang *et al.*, 2022*a*; Jia *et al.*, 2023). The response mechanism of producing high luminescent products after ligand addition has also been revealed. Pyridine is widely used in the manufacturing industry, not only as a raw material but also as a solvent. However, due to its volatility, it is easy for pyridine to enter the atmosphere in the application process, which can cause environmental pollution (Kim & Kim, 2016). Because pyridine is harmful to human health, the detection of pyridine in air is important (Campos *et al.*, 2015). Herein, we report the successfully designed synthesis, crystal structure and properties of a new $Cu_2I_2P_2S_2$-type binuclear complex, namely, di-μ-iodido-bis[(thiourea-κS)(triphenylphosphine-κP)-copper(I)], $Cu_2I_2TPP_2TU_2$ (**I**), where TPP and TU represent the cheap ligands triphenylphosphine and thiourea, respectively (Scheme 1). After composite processing, the paper-based film based on complex **I** exhibits obvious luminescence light-up sensing for pyridine and 4-methylpyridine.



I

Scheme 1

2. Experimental

All reagents were purchased commercially and used without further purification. The elemental analysis of C, H, N and S was performed on an Elementar vario MICRO analytical instrument. The IR spectrum was measured with a Bruker Tensor27 FT–IR spectrophotometer in the range 4000–400 cm^{-1} in KBr lamella. The UV–Vis and photoluminescence spectra of the crystalline powder samples were recorded with Shimadzu UV3600 and Horiba Jobin-Yvon FL3-211-P spectrometers at room temperature, respectively. The sensing spectra of the VOCs (volatile organic components) were measured on a NOVA high sensitive spectrometer (ideaoptics, China).

2.1. Synthesis and crystallization

Crystals of complex **I** was obtained by solution reaction and cooling crystallization. Firstly, the synthetic raw materials of CuI (0.5 mmol, 97 mg), triphenylphosphine (TPP; 0.5 mmol, 132 mg) and thiourea (TU; 0.5 mmol, 40 mg) were added to CH_3CN (40 ml). The mixture was heated to reflux and stirred continuously for 4 h; the reaction solution gradually changed from turbid to clear. After filtration, the solution was placed in a refrigerator (–4 °C) for crystallization. A few days later, a large number of colourless block-shaped crystals of **I** were obtained (yield 41.3%). IR (KBr, cm^{-1}): 3413 (*s*), 3294 (*s*), 3186 (*ms*), 3053 (*m*), 2999 (*w*), 2679 (*w*), 1959 (*w*), 1890 (*w*), 1821 (*w*), 1604 (*s*), 1480 (*ms*), 1437 (*ms*), 1404 (*vs*), 1096 (*ms*), 1022 (*w*), 746 (*s*), 702 (*s*), 514 (*s*), 475 (*ms*). Analysis calculated (%) for $C_{38}H_{38}Cu_2I_2N_4P_2S_2$: C 43.15, H 3.62, N 5.30, S 6.06; found: C 43.32, H 3.49, N 5.13, S 6.28. A single crystal suitable for X-ray analysis was selected from these crystals. Additionally, a large number of crystalline powder samples were prepared by the ball-milling method in a yield of 93.15%. The phase of this powder was confirmed by powder X-ray diffraction (PXRD).

Table 1
Experimental details.

Crystal data	
Chemical formula	$[Cu_2I_2(CH_4N_2S)_2(C_{18}H_{15}P)_2]$
M_r	1057.66
Crystal system, space group	Triclinic, $P\overline{1}$
Temperature (K)	170
a, b, c (Å)	8.7405 (8), 9.3363 (9), 14.3558 (13)
α, β, γ (°)	89.212 (4), 72.987 (4), 64.826 (3)
V (Å^3)	1005.50 (16)
Z	1
Radiation type	Mo $K\alpha$
μ (mm^{-1})	2.81
Crystal size (mm)	0.49 × 0.29 × 0.23
Data collection	
Diffractometer	Bruker APEXII CCD
Absorption correction	Multi-scan (*SADABS*; Bruker, 2016)
T_{min}, T_{max}	0.423, 0.746
No. of measured, independent and observed [$I > 2\sigma(I)$] reflections	17531, 3947, 3709
R_{int}	0.052
$(\sin \theta/\lambda)_{max}$ (Å^{-1})	0.616
Refinement	
$R[F^2 > 2\sigma(F^2)]$, $wR(F^2)$, S	0.041, 0.105, 1.08
No. of reflections	3947
No. of parameters	230
No. of restraints	6
H-atom treatment	H-atom parameters constrained
$\Delta\rho_{max}$, $\Delta\rho_{min}$ (e Å^{-3})	2.32, −1.76

computer programs: *APEX2* (Bruker, 2016), *SAINT* (Bruker, 2016), *SHELXS* (Sheldrick, 2008), *SHELXL2018* (Sheldrick, 2015) and *OLEX2* (Dolomanov *et al.*, 2009).

2.2. Refinement

Crystal data, data collection and structure refinement details are summarized in Table 1. All H atoms bonded to C and N atoms were added at calculated positions and refined using a riding model, with C—H = 0.95 Å and $U_{iso}(H) = 1.2U_{eq}(C)$, and N—H = 0.88 Å and $U_{iso}(H) = 1.2U_{eq}(N)$. The thiourea ligands were modelled as disordered over two sites. Appropriate restraints and constraints were applied to this disordered group to ensure that the ligand's geometric and displacement parameters approximated to normal behaviour.

3. Results and discussion

Single-crystal X-ray diffraction analysis revealed that complex **I** is a centrosymmetric binuclear cluster structure, in which the Cu^I centre is coordinated by two μ_2-I, one TPP and one TU

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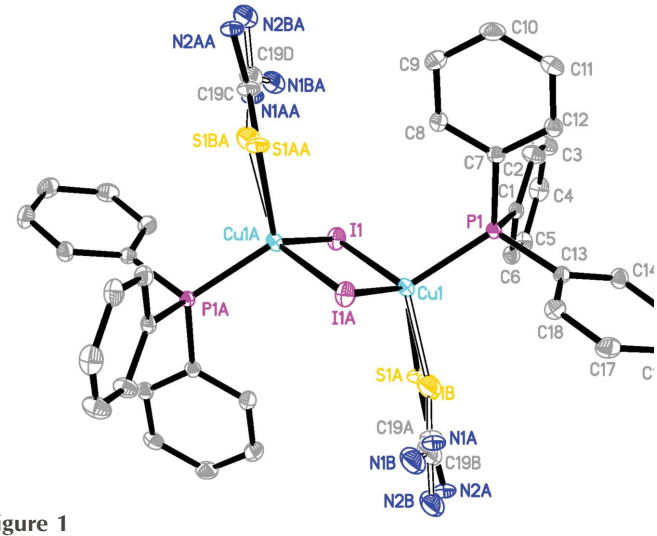

Figure 1
The molecular structure and labelling of title complex **I**, with displacement ellipsoids drawn at the 30% probability level and H atoms omitted for clarity.

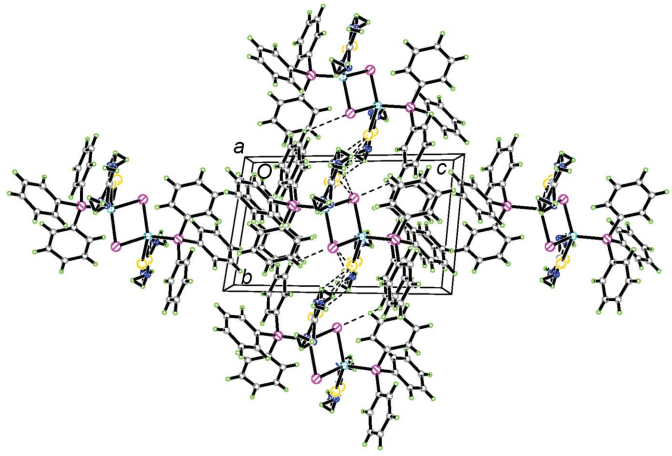

Figure 2
The packing diagram of title complex **I**, viewed along the *a* direction.

Table 2
Selected bond lengths (Å).

I1—Cu1^{i}	2.6189 (7)	Cu1—S1*A*	2.3005 (16)
I1—Cu1	2.7147 (6)	Cu1—P1	2.2510 (11)
Cu1—Cu1^{i}	2.9792 (11)	Cu1—S1*B*	2.348 (6)

Symmetry code: (i) $-x, -y+1, -z+1$.

Table 3
Hydrogen-bond geometry (Å, °).

D—H···*A*	*D*—H	H···*A*	*D*···*A*	*D*—H···*A*
N1—H1*B*···I1^{i}	0.86	2.93	3.7738 (4)	166
N2—H2*A*···S1*A*^{ii}	0.86	2.76	3.3753 (3)	129
N2—H2*B*···I1^{iii}	0.86	2.91	3.7094 (4)	153

Symmetry codes: (i) $-x, -y+1, -z+1$; (ii) $-x+1, -y, -z+1$; (iii) $x+1, y, z$.

ligand (Fig. 1). In the binuclear cluster, two CuI_2PS tetrahedra are fused together in an edge-shared fashion by two μ_2-I ligands. The $Cu_2I_2P_2S_2$ cluster core is quite regular, the CuI_2Cu quadrilateral is completely in the same plane and the P—Cu—Cu—P fragment is the same. The dihedral angle between these two planes is also very close to perpendicular [95.62 (5)°]. According to a search in the Cambridge Structural Database (CSD, Version 5.43; Groom *et al.*, 2016), only 11 structures display the $Cu_2I_2P_2S_2$ cluster structure in numerous CuI–TPP complexes (Lobana *et al.*, 2006, 2007, 2009*a*,*b*,*c*, 2010, 2012), with the $Cu_2I_2P_2N_2$ or $Cu_2I_2P_{2-4}$ cluster structure being relatively more common (Tsuge *et al.*, 2016; Troyano *et al.*, 2021; Dai *et al.*, 2023). In the cluster, the Cu—I, Cu—S and Cu—P bond lengths (listed in Table 2) are all in the normal ranges (Xu *et al.*, 2014; Wang *et al.*, 2020*a*,*c*; Zhu *et al.*, 2018; Bonnot *et al.*, 2016; Knorr *et al.*, 2015; Liang *et al.*, 2021). In this structure, the TU ligand only utilizes the S atom in a terminal coordination fashion, while its amino N atoms are unemployed (Lobana *et al.*, 2008; Wang *et al.*, 2020*b*). Although the amino group often serves as a donor of a hydrogen bond (He *et al.*, 2012), no classic hydrogen-bonding interaction is found. Indeed, *PLATON* (Spek, 2020) checks show that some weak hydrogen-bonding interactions are found (Table 3). These hydrogen-bonding interactions are useful for constructing the supramolecular structure (Tao *et al.*, 2018; Shou *et al.*, 2017*b*; Lin *et al.*, 2015; Zhou *et al.*, 2013).

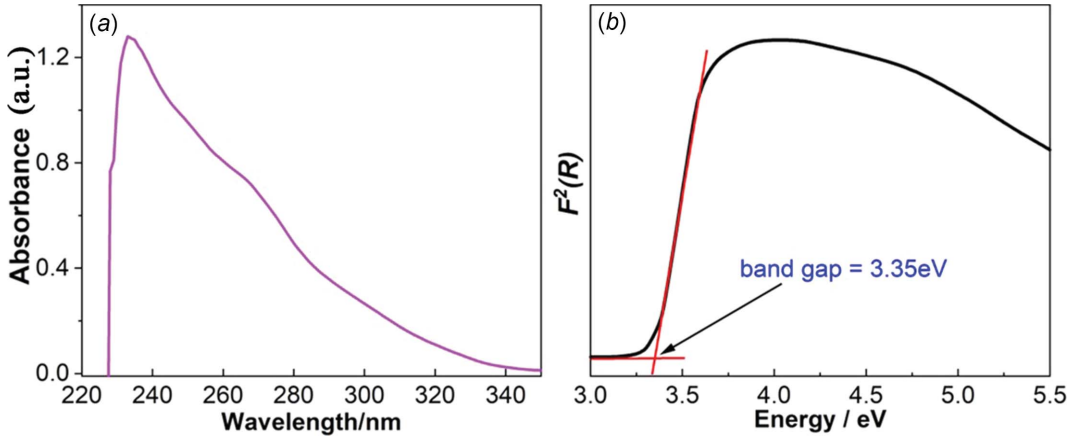

Figure 3
(*a*) The UV absorption spectrum of **I** recorded on a solution sample and (*b*) the UV diffuse reflectance spectrum recorded on a powder sample.

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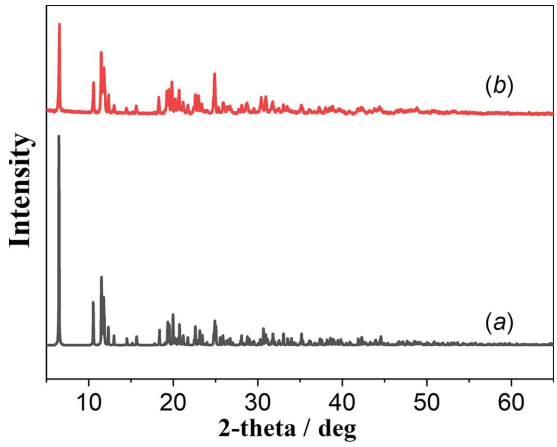

Figure 4
The powder XRD pattern for complex **I**, showing (*a*) the pattern calculated from the single-crystal data and (*b*) the experimental pattern recorded at room temperature.

The packing diagram of this structure is shown in Fig. 2, with some weak hydrogen-bonding interactions.

The UV–Vis absorption and diffuse reflectance spectra of complex **I** recorded on solution and crystalline powder samples are shown in Figs. 3(*a*) and 3(*b*), respectively. The phase of the powder sample has been pre-confirmed by powder X-ray diffraction examination (Fig. 4). As shown in Fig. 4(*a*), only one broad absorption band is seen from about 230 to 340 nm, which should be assigned as the ligand-centred (π–π^*) transition of TPP and the thione ligand (Shou *et al.*, 2017*a*; Zhang *et al.*, 2014; Hong *et al.*, 2014). In addition, there are some weak absorption tails in the longwave region, which should be attributed to the charge-transfer absorption band. According to the previously reported CuXP complexes, the low-energy transition (around 400 nm) is usually assigned to the charge-transfer mechanism of the metal–mixed halide to ligand [(X+M)LCT] transition (Zhu *et al.*, 2018; Shou *et al.*, 2017*a*; Zhang *et al.*, 2014; Hong *et al.*, 2014). From the spectrum of the solid sample, it can be seen that the charge-transfer absorption band in the solid state is stronger. Based on this diffuse reflectance spectrum, the optical band gap of the complex can also be calculated to be 3.35 eV according to the Kubelka–Munk function (Jiang *et al.*, 2019; Wang *et al.*, 2021), belonging to a wide-bandgap semiconductor.

The solid-state emission and excitation spectra have been recorded on crystalline powder samples for complex **I** at room temperature (Fig. 5). This complex emits bluish–green phosphorescence with a lifetime of ~1.86 µs (Fig. S2 in the supporting information). The photoluminescence (PL) curve is a broad band with no obvious vibronic structure. The excitation and emission spectra have maximum values at 365 and 490 nm, respectively. Previously, in other halides or heteroleptic Cu^I complexes, the phosphorescent emission of the lowest triplet state (T_1) was usually assigned to some (X+M)LCT or (L′+M)LCT excitation (Zhang *et al.*, 2015; Czerwieniec *et al.*, 2011; Chen *et al.*, 2013; Xu *et al.*, 2014; Chai *et al.*, 2015; Shou *et al.*, 2017*b*; Tao *et al.*, 2018). Here, its PL emission should also be assigned to some charge-transfer excited state.

Using a paper-based sensing film process, it was found that complex **I** exhibited an obvious fluorescent lighting response for pyridine and 4-methylpyridine. The paper-based sensing film was made according to a method reported previously (Wang *et al.*, 2022*a*,*b*; Zhang *et al.*, 2023; Tang *et al.*, 2023; Jia *et al.*, 2023; Dai *et al.*, 2023). The luminescence of the paper-based film itself is not strong. When the paper-based film was exposed to pyridine or 4-methylpyridine vapour, its PL (photoluminescent) emission showed a rapid enhancement phenomenon (Figs. 6 and 7). As shown in Fig. 6, from the measurement and calculation of initial exposure, the fluorescence intensity at 10 and 30 s after exposure reached about 2.4 and 3.6 times the initial value, respectively. The T_{50}/T_{90} is about 7/45 s; the so-called T_{50}/T_{90} refers to the time required for the gas concentration detector to change from reading 0 to 50/90%. Therefore, this paper-based sensor can be used as a convenient and very rapid response tool for pyridine vapour detection. For 4-methylpyridine, the magnification of PL

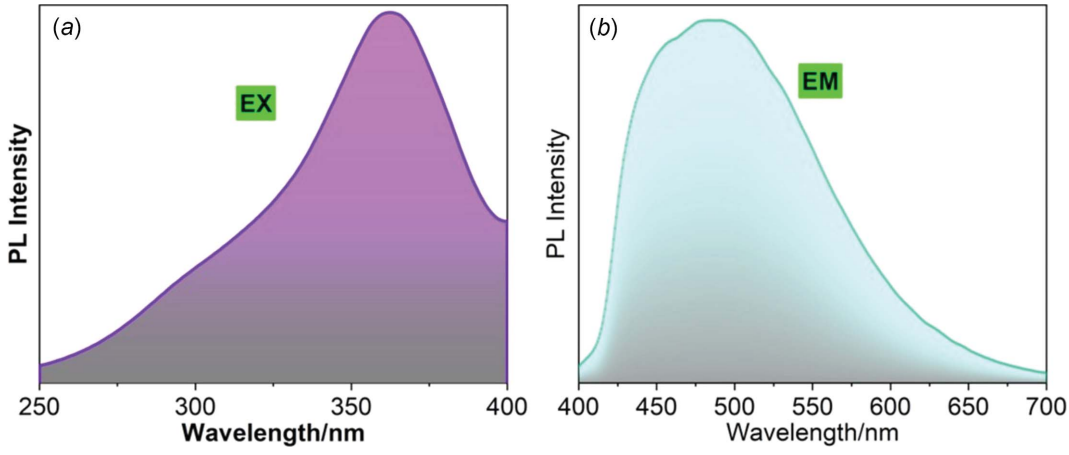

Figure 5
The photoluminescent (*a*) excitation (λ_{em} = 490 nm) and (*b*) emission (λ_{ex} = 365 nm) spectra recorded on a powder sample of complex **I** at room temperature.

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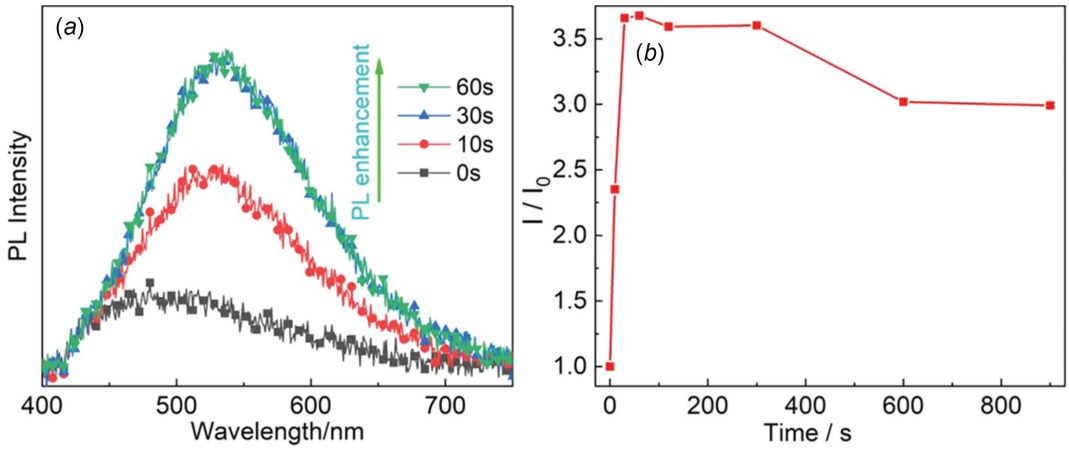

Figure 6
PL enhanced response of paper-based sensor **1** to pyridine: (*a*) a series of PL emission curves illustrate the enhancement effect (λ_{ex} = 365 nm) and (*b*) correlation curve between fluorescence intensity and exposure time in pyridine.

enhancement is much greater (~8 times), but its response rate is much slower. Its T_{50}/T_{90} is about 30/300 s. Based on the previous research results (Dai *et al.*, 2023; Jia *et al.*, 2023), a reasonable sensing mechanism is proposed. The proposed mechanism is that fluorescence sensing is derived from the coordination reaction of pyridine vapour with the tricoordinated Cu centre of the complex, which not only changes the coordination geometry of Cu^I, but also changes the molecular orbital and excited-state characteristics, thereby changing the luminescence properties. Here, the sensing mechanism should also be similar, as pyridine replaces the position of thiourea to produce high fluorescence responsive products.

4. Conclusion

In summary, a $Cu_2I_2P_2S_2$-type binuclear complex, Cu_2I_2-TPP_2TU_2 (**I**) was rationally synthesized from the cheap ligands TPP and TU and characterized by single-crystal X-ray crystallography, as well as by elemental analysis and UV–Vis and photoluminescence spectroscopies. In this complex, each Cu^I centre adopts a CuI_2PS four-coordination mode and pairs of centres are connected by two μ_2-I ligands to form a centrosymmetric binuclear cluster. It was also found that the paper-based film of this complex exhibited obvious luminescence light-up sensing for pyridine and 4-methylpyridine. PL sensing for pyridine is vast; the T_{50}/T_{90} is as low as 7/45 s. Although the PL sensing for 4-methylpyridine is slightly slow, the lighting-up is very significant, with a PL enhancement ratio of 8 times.

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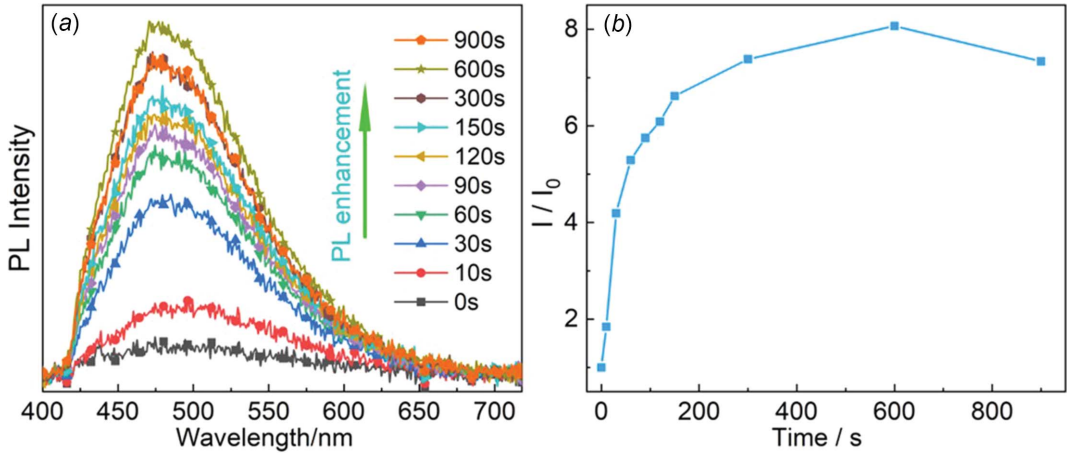

Figure 7
PL enhanced response of paper-based sensor **1** to 4-methylpyridine: (*a*) a series of PL emission curves illustrate the enhancement effect (λ_{ex} = 365 nm) and (*b*) correlation curve between fluorescence intensity and exposure time in 4-methylpyridine.

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supporting information

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A luminescent $Cu_2I_2P_2S_2$-type binuclear complex and its fluorescence sensing for pyridine

Jin-Tao Wu, Li Song, Wen-Ze Xu, Xin-Yu Wei, Yu-Xin Zhang, Ying-Ying Zhang, Xin-Yang Du and Wen-Xiang Chai

Computing details

Di-μ-iodido-bis[(thiourea-κS)(triphenylphosphine-κP)copper(I)]

Crystal data

$[Cu_2I_2(CH_4N_2S)_2(C_{18}H_{15}P)_2]$
M_r = 1057.66
Triclinic, $P\overline{1}$
a = 8.7405 (8) Å
b = 9.3363 (9) Å
c = 14.3558 (13) Å
α = 89.212 (4)°
β = 72.987 (4)°
γ = 64.826 (3)°
V = 1005.50 (16) Å^3
Z = 1
F(000) = 520
D_x = 1.747 Mg m^{-3}
Mo $K\alpha$ radiation, λ = 0.71073 Å
Cell parameters from 9860 reflections
θ = 2.4–27.1°
μ = 2.81 mm^{-1}
T = 170 K
Block, colourless
0.49 × 0.29 × 0.23 mm

Data collection

Bruker APEXII CCD diffractometer
φ and ω scans
Absorption correction: multi-scan (SADABS; Bruker, 2016)
T_{min} = 0.423, T_{max} = 0.746
17531 measured reflections
3947 independent reflections
3709 reflections with $I > 2\sigma(I)$
R_{int} = 0.052
θ_{max} = 26.0°, θ_{min} = 2.4°
$h = -9 \rightarrow 10$
$k = -11 \rightarrow 11$
$l = -17 \rightarrow 17$

Refinement

Refinement on F^2
Least-squares matrix: full
$R[F^2 > 2\sigma(F^2)]$ = 0.041
$wR(F^2)$ = 0.105
S = 1.08
3947 reflections
230 parameters
6 restraints
Primary atom site location: structure-invariant direct methods
Hydrogen site location: mixed
H-atom parameters constrained
$w = 1/[\sigma^2(F_o^2) + (0.035P)^2 + 4.5581P]$ where $P = (F_o^2 + 2F_c^2)/3$
$(\Delta/\sigma)_{max}$ = 0.001
$\Delta\rho_{max}$ = 2.32 e Å^{-3}
$\Delta\rho_{min}$ = −1.76 e Å^{-3}

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Special details

Geometry. All esds (except the esd in the dihedral angle between two l.s. planes) are estimated using the full covariance matrix. The cell esds are taken into account individually in the estimation of esds in distances, angles and torsion angles; correlations between esds in cell parameters are only used when they are defined by crystal symmetry. An approximate (isotropic) treatment of cell esds is used for estimating esds involving l.s. planes.

Fractional atomic coordinates and isotropic or equivalent isotropic displacement parameters (Å²)

	x	*y*	*z*	U_{iso}*/U_{eq}	Occ. (<1)
I1	1.89594 (4)	−1.67987 (4)	0.53274 (2)	0.03301 (12)	
Cu1	2.14519 (7)	−1.60348 (7)	0.41096 (4)	0.02722 (15)	
S1A	2.3792 (4)	−1.8210 (4)	0.4361 (3)	0.0269 (9)	0.579 (15)
P1	2.12866 (14)	−1.62692 (12)	0.25878 (7)	0.0191 (2)	
N2A	2.7090 (10)	−1.9137 (9)	0.4251 (6)	0.0269 (9)	0.579 (15)
H2AA	2.802733	−1.894665	0.413626	0.032*	0.579 (15)
H2AB	2.717488	−2.008669	0.438245	0.032*	0.579 (15)
N1A	2.5694 (9)	−1.6568 (8)	0.4005 (7)	0.0269 (9)	0.579 (15)
H1AA	2.671431	−1.679966	0.354259	0.032*	0.579 (15)
H1AB	2.480732	−1.595508	0.379292	0.032*	0.579 (15)
C13	2.3362 (6)	−1.6591 (5)	0.1645 (3)	0.0219 (8)	
C19A	2.5542 (11)	−1.7988 (9)	0.4227 (8)	0.0269 (9)	0.579 (15)
C1	2.0928 (6)	−1.7995 (5)	0.2332 (3)	0.0212 (8)	
C6	2.1924 (6)	−1.9419 (5)	0.2644 (3)	0.0268 (9)	
H6	2.277435	−1.946584	0.294528	0.032*	
C7	1.9588 (6)	−1.4609 (5)	0.2233 (3)	0.0214 (8)	
C12	1.9797 (6)	−1.4258 (5)	0.1265 (3)	0.0250 (9)	
H12	2.085452	−1.491225	0.075253	0.030*	
C2	1.9682 (7)	−1.7954 (6)	0.1899 (4)	0.0337 (11)	
H2	1.900144	−1.699698	0.167849	0.040*	
C8	1.8019 (6)	−1.3644 (6)	0.2966 (3)	0.0319 (10)	
H8	1.786686	−1.386451	0.362664	0.038*	
C14	2.4228 (6)	−1.7786 (6)	0.0837 (3)	0.0284 (9)	
H14	2.371505	−1.846827	0.074776	0.034*	
C18	2.4141 (6)	−1.5605 (6)	0.1755 (4)	0.0296 (10)	
H18	2.357487	−1.479575	0.230142	0.035*	
C5	2.1671 (7)	−2.0770 (6)	0.2514 (3)	0.0322 (10)	
H5	2.237676	−2.174324	0.270900	0.039*	
C4	2.0408 (8)	−2.0701 (6)	0.2104 (4)	0.0388 (12)	
H4	2.021077	−2.161495	0.204143	0.047*	
C17	2.5740 (7)	−1.5791 (6)	0.1074 (4)	0.0354 (11)	
H17	2.625099	−1.509871	0.114788	0.043*	
C10	1.6897 (7)	−1.2014 (6)	0.1795 (4)	0.0383 (12)	
H10	1.598134	−1.112909	0.164460	0.046*	
C16	2.6582 (7)	−1.6994 (6)	0.0288 (4)	0.0368 (11)	
H16	2.768594	−1.714044	−0.017190	0.044*	
C11	1.8462 (7)	−1.2956 (6)	0.1054 (4)	0.0325 (10)	
H11	1.861817	−1.270655	0.039865	0.039*	
C3	1.9424 (8)	−1.9309 (7)	0.1785 (4)	0.0419 (13)	

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H3	1.857192	−1.927253	0.148762	0.050*	
C15	2.5825 (7)	−1.7973 (6)	0.0171 (4)	0.0363 (11)	
H15	2.640540	−1.878729	−0.037348	0.044*	
C9	1.6669 (7)	−1.2361 (6)	0.2747 (4)	0.0414 (13)	
H9	1.559164	−1.172396	0.325239	0.050*	
S1B	2.4207 (13)	−1.7895 (11)	0.4153 (6)	0.067 (2)	0.421 (15)
C19B	2.589 (2)	−1.7827 (19)	0.4257 (15)	0.067 (2)	0.421 (15)
N2B	2.723 (2)	−1.906 (2)	0.4477 (12)	0.067 (2)	0.421 (15)
H2BA	2.788908	−1.871994	0.470306	0.080*	0.421 (15)
H2BB	2.681590	−1.958857	0.493506	0.080*	0.421 (15)
N1B	2.583 (2)	−1.6260 (17)	0.4327 (11)	0.067 (2)	0.421 (15)
H1BA	2.548541	−1.574358	0.382895	0.080*	0.421 (15)
H1BB	2.693817	−1.636700	0.427860	0.080*	0.421 (15)

Atomic displacement parameters (Å²)

	U^{11}	U^{22}	U^{33}	U^{12}	U^{13}	U^{23}
I1	0.0382 (2)	0.02971 (18)	0.02997 (18)	−0.01769 (14)	−0.00495 (13)	0.00112 (12)
Cu1	0.0290 (3)	0.0319 (3)	0.0219 (3)	−0.0123 (2)	−0.0112 (2)	0.0015 (2)
S1A	0.0208 (12)	0.0247 (12)	0.0410 (15)	−0.0115 (8)	−0.0157 (9)	0.0070 (8)
P1	0.0200 (5)	0.0186 (5)	0.0181 (5)	−0.0076 (4)	−0.0063 (4)	0.0001 (4)
N2A	0.0208 (12)	0.0247 (12)	0.0410 (15)	−0.0115 (8)	−0.0157 (9)	0.0070 (8)
N1A	0.0208 (12)	0.0247 (12)	0.0410 (15)	−0.0115 (8)	−0.0157 (9)	0.0070 (8)
C13	0.022 (2)	0.0209 (19)	0.0211 (19)	−0.0070 (16)	−0.0076 (16)	0.0035 (16)
C19A	0.0208 (12)	0.0247 (12)	0.0410 (15)	−0.0115 (8)	−0.0157 (9)	0.0070 (8)
C1	0.024 (2)	0.0196 (19)	0.0196 (19)	−0.0105 (17)	−0.0043 (16)	−0.0008 (15)
C6	0.030 (2)	0.024 (2)	0.025 (2)	−0.0106 (18)	−0.0080 (18)	0.0015 (17)
C7	0.022 (2)	0.0198 (19)	0.023 (2)	−0.0092 (17)	−0.0088 (17)	0.0011 (16)
C12	0.028 (2)	0.023 (2)	0.025 (2)	−0.0110 (18)	−0.0089 (18)	−0.0004 (16)
C2	0.038 (3)	0.029 (2)	0.044 (3)	−0.018 (2)	−0.023 (2)	0.008 (2)
C8	0.028 (2)	0.034 (2)	0.025 (2)	−0.006 (2)	−0.0074 (19)	−0.0004 (19)
C14	0.031 (2)	0.029 (2)	0.026 (2)	−0.0142 (19)	−0.0073 (19)	−0.0005 (18)
C18	0.029 (2)	0.028 (2)	0.033 (2)	−0.0129 (19)	−0.0090 (19)	0.0016 (19)
C5	0.042 (3)	0.022 (2)	0.028 (2)	−0.013 (2)	−0.005 (2)	0.0025 (17)
C4	0.050 (3)	0.032 (3)	0.040 (3)	−0.029 (2)	−0.005 (2)	0.000 (2)
C17	0.033 (3)	0.039 (3)	0.045 (3)	−0.022 (2)	−0.016 (2)	0.013 (2)
C10	0.034 (3)	0.029 (2)	0.047 (3)	−0.003 (2)	−0.023 (2)	0.004 (2)
C16	0.027 (2)	0.043 (3)	0.035 (3)	−0.016 (2)	−0.003 (2)	0.009 (2)
C11	0.038 (3)	0.033 (2)	0.033 (2)	−0.016 (2)	−0.021 (2)	0.009 (2)
C3	0.048 (3)	0.043 (3)	0.056 (3)	−0.031 (3)	−0.029 (3)	0.009 (3)
C15	0.033 (3)	0.040 (3)	0.026 (2)	−0.014 (2)	0.001 (2)	−0.004 (2)
C9	0.029 (3)	0.038 (3)	0.039 (3)	0.002 (2)	−0.011 (2)	−0.004 (2)
S1B	0.061 (4)	0.065 (3)	0.049 (3)	0.004 (3)	−0.032 (2)	−0.010 (2)
C19B	0.061 (4)	0.065 (3)	0.049 (3)	0.004 (3)	−0.032 (2)	−0.010 (2)
N2B	0.061 (4)	0.065 (3)	0.049 (3)	0.004 (3)	−0.032 (2)	−0.010 (2)
N1B	0.061 (4)	0.065 (3)	0.049 (3)	0.004 (3)	−0.032 (2)	−0.010 (2)

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Geometric parameters (Å, º)

I1—Cu1^{i}	2.6192 (7)	C8—H8	0.9500
I1—Cu1	2.7147 (7)	C8—C9	1.390 (7)
Cu1—Cu1^{i}	2.9794 (11)	C14—H14	0.9500
Cu1—S1A	2.306 (3)	C14—C15	1.382 (7)
Cu1—P1	2.2509 (12)	C18—H18	0.9500
Cu1—S1B	2.313 (6)	C18—C17	1.392 (7)
S1A—C19A	1.586 (7)	C5—H5	0.9500
P1—C13	1.823 (4)	C5—C4	1.374 (8)
P1—C1	1.828 (4)	C4—H4	0.9500
P1—C7	1.824 (4)	C4—C3	1.379 (8)
N2A—H2AA	0.8800	C17—H17	0.9500
N2A—H2AB	0.8800	C17—C16	1.387 (8)
N2A—C19A	1.332 (9)	C10—H10	0.9500
N1A—H1AA	0.8814	C10—C11	1.388 (7)
N1A—H1AB	0.8815	C10—C9	1.377 (8)
N1A—C19A	1.411 (9)	C16—H16	0.9500
C13—C14	1.402 (6)	C16—C15	1.373 (8)
C13—C18	1.391 (6)	C11—H11	0.9500
C1—C6	1.398 (6)	C3—H3	0.9500
C1—C2	1.391 (7)	C15—H15	0.9500
C6—H6	0.9500	C9—H9	0.9500
C6—C5	1.393 (7)	S1B—C19B	1.545 (12)
C7—C12	1.401 (6)	C19B—N2B	1.365 (14)
C7—C8	1.390 (6)	C19B—N1B	1.448 (15)
C12—H12	0.9500	N2B—H2BA	0.8996
C12—C11	1.386 (7)	N2B—H2BB	0.8988
C2—H2	0.9500	N1B—H1BA	0.9101
C2—C3	1.395 (7)	N1B—H1BB	0.9101
Cu1^{i}—I1—Cu1	67.89 (2)	C3—C2—H2	119.7
I1^{i}—Cu1—I1	112.11 (2)	C7—C8—H8	119.6
I1^{i}—Cu1—Cu1^{i}	57.58 (2)	C7—C8—C9	120.9 (4)
I1—Cu1—Cu1^{i}	54.53 (2)	C9—C8—H8	119.6
S1A—Cu1—I1^{i}	117.28 (8)	C13—C14—H14	119.9
S1A—Cu1—I1	93.22 (9)	C15—C14—C13	120.2 (5)
S1A—Cu1—Cu1^{i}	117.01 (11)	C15—C14—H14	119.9
P1—Cu1—I1^{i}	113.05 (3)	C13—C18—H18	119.6
P1—Cu1—I1	106.81 (4)	C13—C18—C17	120.8 (4)
P1—Cu1—Cu1^{i}	127.42 (4)	C17—C18—H18	119.6
P1—Cu1—S1A	112.28 (11)	C6—C5—H5	119.8
P1—Cu1—S1B	109.21 (15)	C4—C5—C6	120.3 (5)
S1B—Cu1—I1	107.3 (4)	C4—C5—H5	119.8
S1B—Cu1—Cu1^{i}	123.05 (16)	C5—C4—H4	119.9
C19A—S1A—Cu1	115.1 (3)	C5—C4—C3	120.2 (5)
C13—P1—Cu1	111.75 (15)	C3—C4—H4	119.9
C13—P1—C1	104.43 (19)	C18—C17—H17	120.3

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C13—P1—C7	103.78 (19)	C16—C17—C18	119.5 (5)
C1—P1—Cu1	112.80 (14)	C16—C17—H17	120.3
C7—P1—Cu1	118.44 (14)	C11—C10—H10	120.0
C7—P1—C1	104.3 (2)	C9—C10—H10	120.0
H2AA—N2A—H2AB	120.0	C9—C10—C11	120.1 (4)
C19A—N2A—H2AA	120.0	C17—C16—H16	119.9
C19A—N2A—H2AB	120.0	C15—C16—C17	120.2 (5)
H1AA—N1A—H1AB	109.3	C15—C16—H16	119.9
C19A—N1A—H1AA	109.5	C12—C11—C10	120.3 (4)
C19A—N1A—H1AB	108.8	C12—C11—H11	119.8
C14—C13—P1	124.0 (3)	C10—C11—H11	119.8
C18—C13—P1	117.3 (3)	C2—C3—H3	120.0
C18—C13—C14	118.6 (4)	C4—C3—C2	120.0 (5)
N2A—C19A—S1A	124.8 (6)	C4—C3—H3	120.0
N2A—C19A—N1A	110.9 (7)	C14—C15—H15	119.7
N1A—C19A—S1A	124.2 (6)	C16—C15—C14	120.7 (5)
C6—C1—P1	116.7 (3)	C16—C15—H15	119.7
C2—C1—P1	124.5 (3)	C8—C9—H9	120.1
C2—C1—C6	118.8 (4)	C10—C9—C8	119.8 (5)
C1—C6—H6	119.9	C10—C9—H9	120.1
C5—C6—C1	120.2 (4)	C19B—S1B—Cu1	135.4 (7)
C5—C6—H6	119.9	N2B—C19B—S1B	124.5 (13)
C12—C7—P1	123.3 (3)	N2B—C19B—N1B	117.8 (13)
C8—C7—P1	117.9 (3)	N1B—C19B—S1B	116.4 (10)
C8—C7—C12	118.8 (4)	C19B—N2B—H2BA	111.4
C7—C12—H12	120.0	C19B—N2B—H2BB	112.1
C11—C12—C7	120.0 (4)	H2BA—N2B—H2BB	107.6
C11—C12—H12	120.0	C19B—N1B—H1BA	109.3
C1—C2—H2	119.7	C19B—N1B—H1BB	109.1
C1—C2—C3	120.5 (5)	H1BA—N1B—H1BB	109.5

Cu1—S1A—C19A—N2A	−173.7 (8)	C1—P1—C13—C18	−169.1 (3)
Cu1—S1A—C19A—N1A	2.1 (11)	C1—P1—C7—C12	−83.3 (4)
Cu1—P1—C13—C14	131.6 (3)	C1—P1—C7—C8	97.2 (4)
Cu1—P1—C13—C18	−46.8 (4)	C1—C6—C5—C4	−1.9 (7)
Cu1—P1—C1—C6	−42.9 (4)	C1—C2—C3—C4	0.1 (9)
Cu1—P1—C1—C2	133.7 (4)	C6—C1—C2—C3	0.4 (7)
Cu1—P1—C7—C12	150.3 (3)	C6—C5—C4—C3	2.4 (8)
Cu1—P1—C7—C8	−29.2 (4)	C7—P1—C13—C14	−99.6 (4)
Cu1—S1B—C19B—N2B	165.0 (13)	C7—P1—C13—C18	81.9 (4)
Cu1—S1B—C19B—N1B	−2 (2)	C7—P1—C1—C6	−172.8 (3)
P1—C13—C14—C15	−178.2 (4)	C7—P1—C1—C2	3.8 (4)
P1—C13—C18—C17	179.0 (4)	C7—C12—C11—C10	−1.4 (7)
P1—C1—C6—C5	177.3 (3)	C7—C8—C9—C10	−1.4 (9)
P1—C1—C2—C3	−176.1 (4)	C12—C7—C8—C9	0.6 (7)
P1—C7—C12—C11	−178.7 (4)	C2—C1—C6—C5	0.4 (7)
P1—C7—C8—C9	−179.9 (4)	C8—C7—C12—C11	0.8 (7)
C13—P1—C1—C6	78.6 (4)	C14—C13—C18—C17	0.5 (7)

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C13—P1—C1—C2	−104.8 (4)	C18—C13—C14—C15	0.2 (7)
C13—P1—C7—C12	25.8 (4)	C18—C17—C16—C15	1.3 (8)
C13—P1—C7—C8	−153.7 (4)	C5—C4—C3—C2	−1.5 (9)
C13—C14—C15—C16	−0.2 (8)	C17—C16—C15—C14	−0.6 (8)
C13—C18—C17—C16	−1.3 (7)	C11—C10—C9—C8	0.8 (9)
C1—P1—C13—C14	9.4 (4)	C9—C10—C11—C12	0.6 (8)

Symmetry code: (i) −*x*+4, −*y*−3, −*z*+1.

Hydrogen-bond geometry (Å, º)

D—H···*A*	*D*—H	H···*A*	*D*···*A*	*D*—H···*A*
N1—H1B···I1^{ii}	0.86	2.93	3.7738 (4)	166
N2—H2A···S1*A*^{iii}	0.86	2.76	3.3753 (3)	129
N2—H2B···I1^{iv}	0.86	2.91	3.7094 (4)	153

Symmetry codes: (ii) −*x*, −*y*+1, −*z*+1; (iii) −*x*+1, −*y*, −*z*+1; (iv) *x*+1, *y*, *z*.

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